

Patient-reported outcome measures in sports medicine

with NATALIE COLLINS, KAY CROSSLEY and EWA M ROOS

The operation was entirely successful, but the patient succumbed.

Manual of Operative Surgery, published 1887

Sports medicine clinicians likely had their first encounter with patient-reported outcome measures (PROMs) in the research setting, as PROMs are commonly used to capture treatment response. However, PROMs can also be useful in clinical practice. This chapter outlines what PROMs are, why they are important to use in clinical practice and what constitutes a 'good' PROM. It also provides examples of generic PROMs to use in sports medicine patients, which can be used to complement more specific measures outlined in the regional chapters. We emphasise that the key letters in the term 'PROMs' are the first two—P and R—for 'patient-reported.' That is what differentiates PROMs from 'objective' clinical measures such as range of motion or strength.

WHAT ARE PROMS?

PROMs are measures in which the patient provides his or her own perspective of a health condition or treatment. This is particularly useful when the aspect of interest is either impossible or impractical to observe directly, such as pain or high-level sporting function.¹ PROMs can be specific for the region (e.g. knee), condition (e.g. tendinopathy) or intervention (e.g. anterior cruciate ligament reconstruction or exercise). The clinician may be familiar with measures such as the IKDC (International Knee Documentation Committee) for knee function, the Western Ontario and McMaster Universities Arthritis Index (WOMAC) score for osteoarthritis, and the Victorian Institute of Sport Assessment (VISA) score which rates patients who have tendinopathy. All of these are examples of various types of PROMs.

A type of PROM can also provide a generic measure of overall health status (i.e. not 'disease-specific').² PROMs can

measure a variety of constructs, such as pain, symptoms, physical function and quality of life. An example of a very simple, commonly used PROM in clinical practice is asking the patient to rate their pain severity on a scale from zero to ten—a visual analogue scale (VAS). Contemporary PROMs provide more comprehensive insight into a number of aspects that are important to consider in the patient's condition.

WHY IS IT IMPORTANT TO USE APPROPRIATE PROMS IN SPORTS MEDICINE?

Clinicians who work in sports and exercise often use standard clinical tests to evaluate physical aspects, such as joint range of motion, ligament integrity and functional performance. While these provide important information about particular aspects of the patient's condition, they are unable to measure how the patient feels and performs. PROMs complement clinical tests to evaluate the patient's overall health and performance status more comprehensively.

In sports medicine, PROMs are mostly used to evaluate the effectiveness of treatment. However, they may also:

- focus the clinical appointment toward aspects that most concern the patient (i.e. by asking the patient to complete the PROM prior to the appointment)
- build a profile of the patient's health condition over time and quickly identify those who have experienced improvement or deterioration and may require refinements to their training or management³
- identify dimensions that the patient deems to be important in their treatment planning and goal setting

- predict which patients are likely to have a good prognosis or respond to a particular treatment
- decide a patient's readiness for return to activity³
- empower the patient to monitor his or her own health profile over time, by providing benchmarks for the ideal health state (vs peers, normative cohorts)
- screen patients for injury risk
- provide a method of standardising health status reporting, to allow pooling of data from international cohorts (e.g. ACL registries) and detect clinical patterns that may influence prognosis and response to treatment.

CONSIDERATIONS FOR WHAT CONSTITUTES A 'GOOD' PROM FOR USE IN SPORTS MEDICINE

Several factors should be considered before integrating PROMs into clinical practice. This will ensure that implementing the PROM is feasible and it will help to ensure that the PROMs provide relevant, reliable and valid information about the patient and can measure change in the patient's condition over time. Clinicians can obtain this information from a variety of sources, including the PROM's original publication (e.g. journal article, book), online resources (e.g. specific PROM websites such as www.koos.nu) and systematic reviews of PROMs. It is also important to recognise that measurement properties are often population-specific. This means that clinicians should confirm that the PROM has been evaluated in the specific patient cohort relevant to their patient (e.g. ACL-reconstruction versus knee osteoarthritis).

Is the PROM easy to use in a sports medicine setting?

One of the perceived barriers to integrating PROMs into clinical practice is the burden on patients and clinicians, and how this will affect sports medicine services in the clinic, clubrooms, or during competition and travel. At the elite level, patients are accustomed to completing regular log-books to track factors such as their training load, dietary intake and sleep patterns. As such, the inclusion of appropriate PROMs is unlikely to be too burdensome, especially if it provides information that they see to be relevant to their injury or health condition and goals, and is delivered in the same format as their regular log-books (i.e. mobile device or paper diary).

By definition, PROMs should be patient-administered, without input from the clinician or others, as this captures the patient's impression of their health condition and minimises the potential for bias introduced with clinician-completed measures. This is also likely

to minimise the burden on clinicians, as patients can complete the PROMs prior to their appointment or in the waiting room.

Another consideration regarding patient burden is the format and layout of the PROM. Are the items logical and easy to interpret, or does the patient need to read each item carefully to ensure that they provide the correct response? For example, some PROMs reverse the wording or responses to some items and then invert the scoring (e.g. items 4, 8, 12 and 16 of the Tampa Scale for Kinesiophobia).

Although it may be preferable in clinical practice to use PROMs with the shortest completion time, this needs to be considered along with the information that the PROM provides. Some PROMs that take longer to complete may provide substantially more information about different aspects of the patient's condition (e.g. PROMs such as the Knee Injury and Osteoarthritis Outcome Score (KOOS) and Copenhagen Hip And Groin Outcome Score (HAGOS) that provide separate subscales for pain, function and quality of life take 5-10 minutes to complete) than shorter PROMs that only measure one construct (e.g. numerical rating scale for pain taking seconds to complete). The completion time is also important to consider in the context of how frequently the PROM will be administered. If the frequency is at least 3 months apart, longer PROMs are feasible, while shorter PROMs are more feasible if daily, weekly or monthly administration is desired.

The clinician should also consider how burdensome the PROM is to score and interpret in the sports medicine setting. To minimise clinician burden, we recommend that electronic or online data collection platforms be used where possible. This allows clinicians to receive scores in real time using computer algorithms, which enhances their clinical utility.³ Electronic questionnaires have high agreement with their paper-based predecessors.⁴ Furthermore, information needs to be made available to clinicians regarding how to interpret PROM scores, relate them to the patient's condition and use them to make decisions about treatment.³ The availability of a user guide for the PROM will facilitate ease of administration and scoring.

Finally, the accessibility of the PROM should be considered. Many PROMs are open access and freely available for clinical use (e.g. KOOS). However, some PROMs have licensing requirements and fees associated with their use, even in clinical practice (e.g. WOMAC). Given that sports medicine clinicians can manage patients of various cultural and language backgrounds, they should also consider whether a particular PROM is available in languages other than English and whether non-English versions have undergone cross-cultural validation. This ensures that the same meaning of the items and stability

of measurement properties (e.g. reliability, validity, responsiveness) is retained when the PROM is translated into another language.

Does the PROM evaluate dimensions that are relevant for the patient?

In selecting a PROM for use in sports medicine, it is vital that it evaluates specific dimensions, subscales, items or questions that are relevant for the particular patient. The first step in determining relevance of dimensions is to obtain information regarding the patient group that the PROM was intended for. This is generally available in the PROM's original publication. For example, the iHOT-33 was intended for young active patients with hip disorders,⁵ which would be suitable for use in athletic populations. In comparison, the KOOS was intended for use in young, middle-aged and elderly patients with knee injuries and/or osteoarthritis.⁶ Thus, items in the subscale pertaining to function during sport and recreational activities (e.g. running, jumping, twisting/pivoting) may be more relevant to younger active patients than function in activities of daily living (e.g. putting on socks/stockings).

Validity refers to the degree to which a PROM measures the constructs that it intends to measure.⁷ This is an important consideration when making inferences based on PROM scores.¹

Content validity is the degree to which the content of the PROM adequately reflects the constructs to be measured, with respect to their relevance and comprehensiveness.^{1, 7} A PROM has adequate content validity if the developers provide a clear description of the measurement aim, the target population, concepts to be measured and item selection.⁸ Importantly, patients from the target population should have been involved in the development or selection of included items.⁸ Face validity is a component of content validity that is useful for clinicians. This refers to the degree to which, on face value, the items of a PROM look as though they adequately reflect the construct to be measured.⁷ Clinicians should ensure that PROMs demonstrate content validity and that they do not overrule dimensions that patients deem to be important—it is the patient who is the expert in this scenario.

Construct validity refers to how well the PROM measures its intended constructs.² This is typically evaluated by testing hypotheses regarding the expected relationship between the PROM and other instruments. Evidence for convergent construct validity is provided when higher correlations are found between the PROM and existing measures that evaluate similar constructs.⁸ Evidence for divergent construct validity is provided when lower correlations are observed between the

PROM and existing measures that evaluate dissimilar constructs.⁸ For example, in patients who had undergone hip arthroscopy, the pain subscale of the Copenhagen Hip And Groyn Outcome Score (HAGOS) demonstrated larger correlations with the SF-36 bodily pain subscale and smaller correlations with the mental health subscale.⁹

Clinicians may also see reference to another form of validity in the literature, termed criterion validity. This refers to how well the scores of an instrument reflect a 'gold standard'.⁷ Because there is no gold standard measure of patient-reported outcome,¹⁰ criterion validity is not applicable to PROMs.

Do all items within a PROM measure the same construct?

For scoring and interpretation purposes, it is important that all items within a PROM or individual subscales reflect the same dimension.¹¹ The internal consistency measures how well all items within a PROM are correlated and thus measure the same construct.⁸ This is an important consideration for PROMs that use multiple items to measure a single construct (e.g. Tampa Scale for Kinesiophobia), to ensure that all items are likely to respond in a similar way to a change in the patient's condition.

The first step in determining the internal consistency of a PROM is to examine the structural validity. During the development phase of the PROM, specific analyses (principal component analysis or exploratory factor analysis) are used to determine whether the items form one overall dimension (unidimensionality) or more than one dimension (multidimensionality).⁸ For PROMs that consist of a single overall score (e.g. VISA Score), all items should reflect a single construct.¹ For PROMs that consist of several subscales that are scored individually (e.g. HAGOS), several dimensions should be demonstrated.¹ Clinicians should confirm that all items within an overall score or subscale score, as appropriate, reflect only one dimension.

Once the unidimensionality of the PROM or its subscales has been established, Cronbach's alpha can be used to measure the interrelatedness of all items within a (sub)scale.^{7, 8} Cronbach's alpha is considered adequate if it falls between 0.7 and 0.95. A low Cronbach's alpha (<0.7) indicates that the items are not well correlated and that a single summary score is not appropriate.⁸ A very high Cronbach's alpha (>0.95) is also not ideal and suggests that one or more items within the (sub)scale are redundant.⁸

Can the PROM be trusted to detect true change in the patient and be free from error?

To be able to trust and interpret PROM scores, it is important to establish the error associated with the PROM.

This gives clinicians an idea as to whether any change observed in the PROM between repeated administrations reflects a true change in the patient's condition or is due to error inherent in the measure.

The test-retest reliability of a PROM indicates how stable the results are when you measure a patient's unchanged condition on two separate occasions.² For PROMs that are patient-completed, reliability is considered to be intra-rater, as the same person (the patient) completes the PROM on different occasions.⁷ Typically, test-retest reliability is reported using intraclass correlation coefficients (ICC). This indicates the strength of the relationship between the two administrations, from 0 (no relationship) to 1 (the relationship between the repeated administrations can be predicted without error).¹ Interpretation of ICCs differs depending on whether clinicians intend to use the PROM in individual patients or in groups of patients (e.g. sporting teams).

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Test-retest reliability is considered adequate for use in individual patients if the ICC is greater than 0.9,¹² and for use in groups if the ICC is greater than 0.7.⁸

The absolute measurement error or agreement can also be determined. This provides clinicians with a value, expressed in the measurement unit of the specific PROM, which can be used to determine whether a PROM change score is greater than the error of the measure. The standard error of measurement (SEM) estimates one standard deviation of the error associated with one measurement.¹ This is then used to calculate the minimal detectable change (MDC), also known as the smallest detectable change (SDC). The MDC can be determined for individuals ($MDC_{ind} = 1.96 \times \sqrt{2} \times SEM$), reflecting the smallest change occurring within a patient that can be interpreted as 'real' change (at the 0.05 significance level).⁸ It can also be determined for groups (MDC_{group}), by dividing the MDC_{ind} by $\sqrt{n}$ (where n is the number of participants).⁸ In clinical practice, if a patient's change score on a PROM is greater than the MDC_{ind} , clinicians can interpret this as a true change in the patient's condition.

Is the PROM sensitive enough to detect real change in the patient's condition?

Because PROMs are frequently used to evaluate a patient's response to treatment or monitor their condition over time, they must be sufficiently sensitive to be able to detect real changes in a patient's condition. Responsiveness refers to the ability of a PROM to detect change over time in the

measured construct.⁷ This is arguably the most important measurement property, as it gives an indication of how robust the PROM is on all other measurement properties (e.g. validity, reliability). Responsiveness is specific to the population and treatment of interest, so clinicians should seek evidence from studies that test responsiveness in cohorts that have similar characteristics to the specific patient (e.g. age, injury/condition/treatment).

The difficulty for clinicians is the lack of consensus regarding the optimal method of evaluating responsiveness for PROMs. Earlier studies frequently present effect sizes as a measure of responsiveness. However, this is more a reflection of treatment effects (i.e. the magnitude of the change score), rather than whether the PROM can detect change over time (i.e. validity of the change score).¹¹ More recently, the COSMIN group recommended that responsiveness should be evaluated in a similar way to construct validity, by testing predefined hypotheses regarding the expected correlation between change scores of two PROMs that are administered simultaneously or expected differences in changes between 'known' groups.¹³ PROMs may be compared to a 'global rating of change' (GROC) scale, which is an adequate estimate of the magnitude and direction of health status change,¹⁴ or to other measures of the same construct.¹⁵ Thus, responsiveness can be viewed as longitudinal validity.⁸ Clinicians can gain evidence of PROM responsiveness from studies where predefined hypotheses regarding change scores (e.g. effect sizes) are supported. Effect sizes are also useful when comparing PROMs head to head, or when comparing groups that are better, unchanged or worse.

The other component of responsiveness relates to whether the PROM can distinguish clinically important change from measurement error.¹³ This is particularly important for clinical practice, as clinicians need to be confident that changes they are observing in their patient, measured using PROMs, represent meaningful change in the patient's condition. The minimal clinically important difference (MCID), also known as the minimal important change (MIC), is the smallest change in a PROM score that patients consider to be important.¹¹ The MCID provides clinicians with an estimate of the average amount of change in PROM score that patients would typically consider to represent a meaningful improvement.¹

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When interpreting MCIDs, clinicians should ensure that the MCID is larger than the MDC, indicating that the magnitude of important change is greater than the error associated with the PROM.⁸

One consideration with the ability of a PROM to detect change is whether it demonstrates any floor or ceiling effects when used in particular patient groups. Floor effects occur when patients score the lowest possible value on the (sub)scale at baseline measurement, meaning that there is no room for the PROM to detect further deterioration. Ceiling effects occur when patients score the highest possible value, meaning that the PROM is unable to detect improvements in the condition.

Athletic patients may demonstrate ceiling effects on PROMs that were originally designed for use in older, more inactive cohorts. For example, the KOOS subscale for activities of daily living often demonstrates ceiling effects when used in young ACL-injured¹⁶ and reconstructed patients,¹⁷ as the items in this subscale relate to low-level daily activities such as putting on socks. Also, older patients with osteoarthritis may demonstrate floor effects on PROMs designed for younger, more active cohorts. For example, the KOOS subscale for sport and recreation may demonstrate floor effects when used in older patients who have moderate to severe knee osteoarthritis after ACL reconstruction, because they have extreme difficulty participating in running, jumping and twisting activities.

For PROMS that have multiple subscales, it is up to the clinician to decide whether all subscales are relevant to the patient of interest. Since scores are calculated and presented per subscale, individual subscales can be omitted in the population of interest or at specific time points. For example, it may not be relevant to evaluate the KOOS sport/rec subscale 2 weeks after ACL reconstruction. Floor and ceiling effects are considered acceptable if less than 15% of patients score the lowest or highest score, respectively.⁸

SUMMARY

PROMs measure items that matter to patients. They can add valuable information to the assessment and management of patients. Selected PROMs are easy to integrate into the sports medicine setting, evaluate patient-relevant constructs, represent single dimensions and can detect real changes in the patient's condition (beyond a defined measurement error). The reader is directed to Part B chapters for recommendations regarding PROMs for specific body regions, conditions and interventions.

Table 16.1 gives some examples of commonly used generic PROMs, while an example of the clinical use of a PROM is shown in the box below.

Examples of generic PROMs appropriate for use in clinical sports medicine

A number of generic PROMs can be applied across a variety of patient groups (i.e. different conditions or injuries) and interventions. Table 16.1 lists some commonly used generic

PROMs that measure constructs that are applicable for sports medicine patients.

Table 16.1 Commonly used generic PROMs

Pain

Pain Numerical Rating Scale (NRS)¹⁸

- Single item
- Most commonly 11-point scale, anchored with 'no pain' (0) and 'worst imaginable pain' (10)
- Patient selects the number that represents their pain severity over the nominated time period (e.g. last 24 hours, past week)
- Manual scoring, score range 0–10
- Freely available

Function

Patient-specific Functional Scale¹⁹

- 3–5 items, single aggregate score
- Patient nominates up to 5 activities that are difficult because of their injury/condition, then rates the difficulty for each activity on an 11-point NRS (0 = 'unable to perform activity'; 10 = 'able to perform activity at same level as before injury or problem')
- Manual scoring (average of all scores), score range 0–10
- Freely available

continued

Health-related quality of life**SF-36²⁰**

- 36 items across domains of physical functioning, physical role, bodily pain, general health, vitality, social functioning, emotional role and mental health; can also derive two aggregate summary measures: physical component summary and mental component summary
- Scored by recording some items, summing responses and transforming scores to a 0–100 scale (higher score = better health state). Computer scoring algorithms are necessary and freely available
- Normative data available in multiple populations²¹
- Original version available in English and Arabic, free of charge (www.rand.org). Most recent version (developed in 1998) requires licence and is available in more than 170 languages (www.optum.com/optum-outcomes/what-we-do/health-surveys/sf-36v2-health-survey.html)

AQoL-6D²²

- 20 items across 6 domains (independent living, mental health, relationships, senses, coping, pain)
- Scoring algorithms and an online scoring service are available (www.aqol.com.au). Score range –0.04 (health state worse than death) to 0.0 (death) and 1.00 (full health)
- Available in 6 languages
- Normative data available
- Freely available; requires registration for use in studies, but not clinical practice

Health-related quality of life**Pain Catastrophising Scale²³**

- 13 items across 3 subscales (rumination, magnification, helplessness)
- Manual scoring by summing items. Single aggregate score can be calculated (0–52), as well as individual subscale scores. Higher scores represent higher degree of catastrophising
- Freely available in 21 languages (<http://sullivan-painresearch.mcgill.ca/pcs.php>)

Tampa Scale for Kinesiophobia²⁴

- 17 items evaluating kinesiophobia
- Manual scoring by summing all items (after converting individual scores of 4 items). Total score ranges from 17 to 68, where a high score indicates a high degree of kinesiophobia
- Freely available²⁵

Case study: Measuring knee symptoms, function and quality of life after ACL reconstruction

David is a 34-year-old man who underwent an ACL reconstruction (ACL) (hamstring tendon autograft) of his right knee 7 years ago. He returned to social soccer 18 months after surgery. He has recently been experiencing pain and stiffness in his right knee, and has stopped playing soccer and running as he fears that this may increase his pain and any damage in his knee. His family physician ordered X-rays, which revealed mild to moderate osteoarthritis in his right patellofemoral and tibiofemoral joint compartments. David was then referred for physiotherapy management.

In addition to performing his usual clinical assessment, David's physiotherapist administered the KOOS to better

understand David's knee condition. The KOOS was developed for patients with knee injuries and/or osteoarthritis. It consists of five individually scored subscales: pain, symptoms, function in activities of daily living (ADL), function in sport and recreation activities and knee-related quality of life (QoL). The KOOS is reliable and valid in patients with ACL injury and/or knee osteoarthritis.²⁶ The minimal detectable change (MDC), for individuals in adults with mild OA after ACL reconstruction is: pain 20; symptoms 24.9; ADL 14.4; sport/rec 24.9; QoL 20.5 (calculated from SEM data²⁷.)

Figure 16.1a shows David's KOOS scores at baseline (at his first physiotherapy session) compared to population normative data.²⁸ There are marked differences in scores

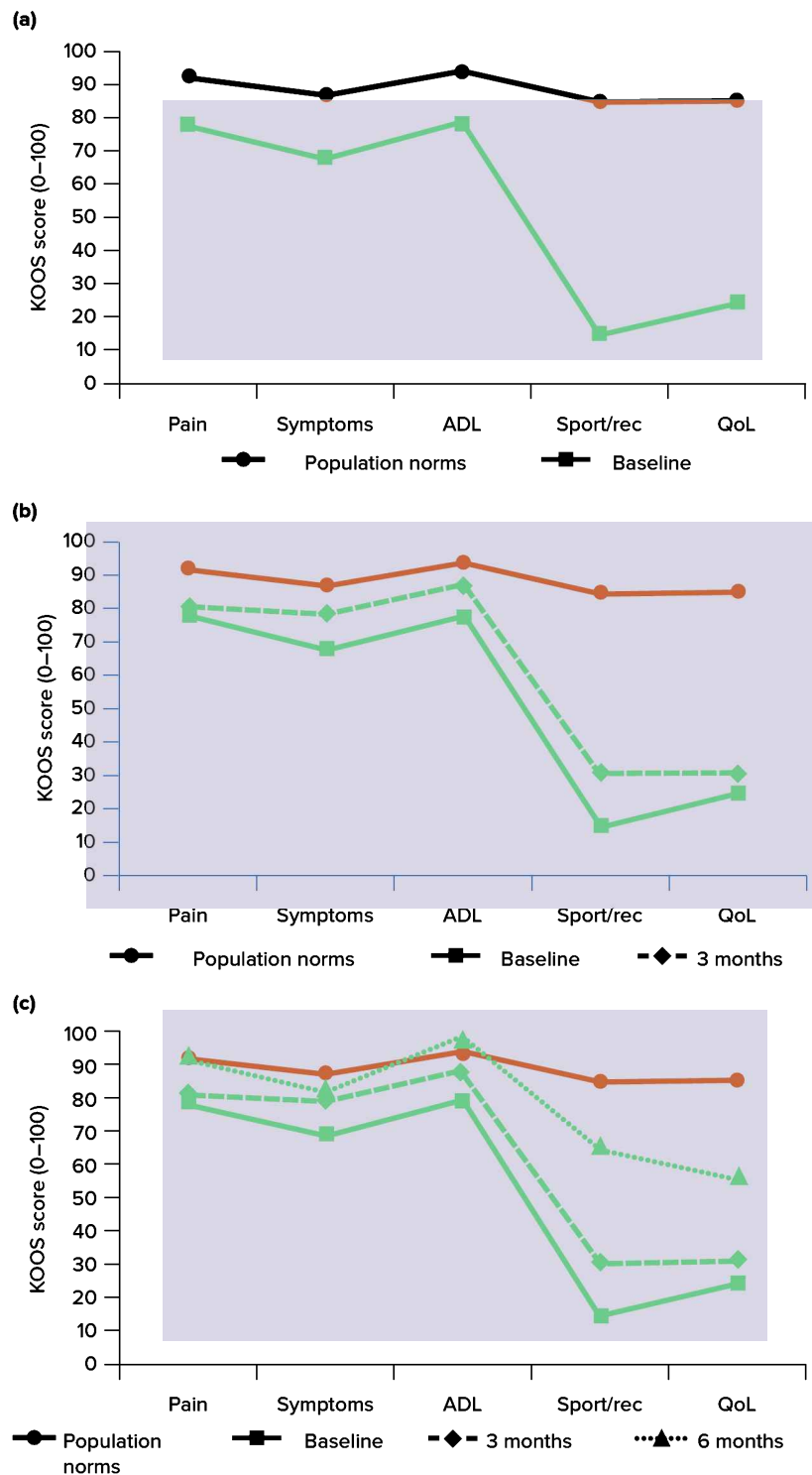


Figure 16.1 David's KOOS profiles of the five subscale scores at (a) baseline, (b) 3 months and (c) 6 months, compared to population normative data

continued

for the sport/rec and QoL subscales. In comparison, the differences in scores for the pain, symptoms and ADL subscales are smaller. This may be because younger patients who undergo ACL reconstruction are more concerned with sport/rec function and QoL and it may reflect ceiling effects observed in this population. However, the difference between David's scores and population norms for the ADL, sport/rec and QoL subscales are greater than the MDC_{ind} , indicating that real differences are present.

The physiotherapist implements a rehabilitation program of lower limb strengthening and neuromuscular control exercises, combined with manual therapy and patella taping to address pain and symptoms.

David's KOOS scores after 3 months are presented in Figure 16.1b. Although his scores are better on the symptoms, ADL and sport/rec subscales, the change scores from baseline are within the error of the measure and therefore may not represent real change. Furthermore, while his scores on the pain, symptoms and ADL subscales are

approaching population norms, the sport/rec and QoL scores remain substantially lower.

Based on these outcomes, the physiotherapist decides to integrate some interventions to address QoL into David's rehabilitation program. Specifically, she uses some graded exercises, combined with education, to improve David's confidence in his knee (assessed by KOOS QoL item 3) and reduce any fear-avoidance of potentially damaging activities (KOOS QoL item 2). She continues to progress his strength and neuromuscular exercises, in preparation for return to sport.

From Figure 16.1c you can see that David's pain, symptoms and ADL scores are almost identical to population norms. His sport/rec and QoL scores have also increased by an amount that exceeds the MDC_{ind} . The physiotherapist interprets these findings as representing a true change in David's knee condition. At the last appointment, the physiotherapist hands over the KOOS profiles to David and asks him to score himself from time to time to keep track of his status.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e